Ousmane Junior SANE

Montpellier, France | myboxe2@gmail.com | 0758845365 | linkedin.com/in/ojsane | github.com/Ousmane-prog

Profile

Engineer in Applied Mathematics and Computational Biology with hands-on experience in scientific computing, modeling and numerical simulation of dynamical systems, as well as robust state estimation for uncertain dynamical systems. Skilled in Python and High Performance Computing with Julia, with the ability to write fast, optimized, maintainable scientific softwares. Experienced in benchmarking, performance profiling and parallel computing.

Interests

Mathematical Modeling, Numerical Simulation, Optimization, Interval Observers, Scientific software development.

Experiences

Scientific Computing Engineer, MISTEA Lab, INRAE – Montpellier, France — Nov 2025 – present

- Developed `IntervalsObservers.jl` for robust state estimation in uncertain dynamical systems.
- Implemented optimal resource allocation strategies for bacterial diauxic growth via `OCP methods`.
- Profiling and benchmarking of neural ODE models to identify computational bottlenecks.

Scientific Software Developer, MICS Lab, CentraleSupélec – Gif-sur-Yvette — Apr 2025 – Aug 2025

- Developed a web application for dose prediction and optimization in MPN patients.
- Implemented Bayesian workflows to fit patient-specific ODE models using Turing.jl.
- Engineered a dose strategy optimization module based on `Hermange et al. (2023)`.

Research Intern in Modeling, Institut de Mathématiques de Marseille (I2M), Marseille — May 2023 – Jul 2023

- Implemented ODE-based mathematical models to simulate heterogeneous tumor growth.
- Qualitative and numerical analysis of the tumor growth model (stability, equilibria, parameter sensitivity).
- Numerical study to identify optimal chemotherapy protocols.

Junior Data Scientist Intern, Institut Laennec, Aix-Marseille Université — Jan 2024 – Jul 2024

- Developed automation scripts for processing and validation of large-scale multicentric clinical datasets.
- Applied univariate and multivariate statistical models for clinical data analysis.
- Collaborated closely with clinicians to interpret results and align analyses with clinical objectives.

Education

2022–2024: Master 2 – Mathématiques appliquées, parcours Computational and Mathematical Biology (CMB) — Aix-Marseille Université

2019–2022: Licence – Mathématiques fondamentales — Aix-Marseille Université

Scientific Computing and Simulation Projects

Interactive SI Epidemic Modeling — github.com/Ousmane-prog

- Interactive Julia dashboard for real-time ODE simulation and Bayesian parameter estimation of an SI epidemic model.
- Tools: Genie.jl, Stipple.jl, DifferentialEquations.jl, Turing.jl, PlotlyBase.jl

Modeling of Intermittent Androgen Suppression Therapy for Prostate Cancer — find report here

- Implemented a hybrid ODE model with discrete therapy switching to compare IAS and CAS.
- Stability analysis (equilibria, Jacobian, eigenvalues) under varying androgen levels.
- IAS shown to extend relapse time by >1 year across three simulated biological scenarios.

Languages

French: native English: fluent

Training

Public Speaking for PhD Students – Academic Writing Center - Centralesupélec Jun 2025
8-hour intensive training focused on scientific communication, presentation techniques, and audience engagement strategies

Centuri Summer School July 2023

- Hands-on Machine Learning: build a classifier to decode neural activity.
- Hands-on data analysis: Analyzed data from neural activity recordings in mice's brain.

Technical Skills

Languages: Julia, Python, C, R, JavaScript